AI & Machine Learning – Task 4

Classification Models, Evaluation Metrics & Handling Imbalanced Data

1. Objective

Task 4 transitions from regression to classification, one of the most widely used problem types in industry (fraud detection, spam filtering, medical diagnosis, churn prediction). The goal is to build a binary classifier, evaluate it using metrics beyond accuracy, handle class imbalance, and justify the final model choice scientifically.

2. Problem & Dataset

Problem: Binary classification — predicting whether a tumor is Malignant (0) or Benign (1).
Dataset: scikit-learn's built-in Breast Cancer dataset (569 samples, 30 features).
Class distribution: 212 Malignant (37.3%) vs 357 Benign (62.7%) — a mild class imbalance, representative of many real-world diagnostic datasets.

3. Step-by-Step Methodology

Steps 1–4: Setup

Libraries imported, dataset loaded, stratified 80/20 train-test split applied (stratify=y preserves class ratio in both sets), and features scaled with StandardScaler.

Step 5: Baseline Model — Logistic Regression

A Logistic Regression classifier (max_iter=1000) was trained as the baseline model.

Step 6: Confusion Matrix & Classification Report

Confusion Matrix: TN=41, FP=1, FN=1, TP=71. A False Negative (predicting Benign when actually Malignant) is the most dangerous error in this context, since it means a cancer diagnosis is missed. Accuracy alone would not reveal this risk — confusion matrix breakdown is essential.

Step 7: Precision, Recall & F1-Score

Precision=0.9861, Recall=0.9861, F1=0.9861. Recall on the malignant/critical class matters most in medical diagnosis: low recall means real disease cases are missed. F1-score is preferred for imbalanced data because it balances precision and recall, preventing a model from hiding poor minority-class performance behind high overall accuracy.

Step 8: ROC Curve & AUC Score

AUC = 0.9954, very close to 1.0, indicating the model separates the two classes almost perfectly across all classification thresholds.

Step 9: Handling Class Imbalance

class_weight='balanced' was applied to Logistic Regression to re-weight classes inversely to their frequency. Result: Accuracy=0.9561, Recall=0.9444, F1=0.9645 — a small trade-off versus baseline since the dataset's imbalance is mild and the baseline already performed very well.

Step 10: Comparison with Decision Tree

A Decision Tree Classifier was trained for comparison: Accuracy=0.9123, Recall=0.9028, F1=0.9286. Train accuracy reached 1.0 vs test accuracy 0.9123 — a gap that signals overfitting, since the unconstrained tree memorized parts of the training data.

4. Results

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954
Logistic Regression (Balanced)	0.9561	0.9855	0.9444	0.9645	—
Decision Tree	0.9123	0.9559	0.9028	0.9286	—

5. Final Model Selection Justification

Selected Model: Baseline Logistic Regression

Metric selection:

Recall on the malignant class and F1-score were prioritized over raw accuracy, since missing a malignant diagnosis (false negative) is far more costly than a false alarm.

Imbalance handling:

The dataset shows mild imbalance (212 vs 357 samples). class_weight='balanced' was tested as a mitigation technique, but the baseline model already achieved strong recall (0.9861) without it, so balancing was not strictly necessary here — though it remains valuable for more severely skewed datasets.

Final decision:

Logistic Regression achieved the best overall results: Accuracy=0.9825, F1=0.9861, AUC=0.9954, with only 1 false positive and 1 false negative in the test set — an excellent, well-balanced outcome.

Trade-offs:

Logistic Regression is more interpretable and stable than the Decision Tree, which overfit the training data (train accuracy 1.0 vs test accuracy 0.9123). The simpler linear model generalized better in this case.

6. Key Learnings

- Accuracy alone can be misleading, especially with imbalanced classes.
- Confusion matrix breakdown (TP/TN/FP/FN) reveals which errors actually matter.
- Precision-recall trade-off must be judged by the real-world cost of each error type.
- ROC-AUC summarizes classifier performance across all probability thresholds.
- class_weight='balanced' is a simple, effective first step for handling imbalance.
- Simpler models can outperform complex ones when data is close to linearly separable.

Appendix: Performance Charts